Quiz 4: MA 5755, Department of Mathematics, IIT Madras

Course name : $DataAnalysis\&VisualizationinR/Python/SQL$ **Max. Marks: 25**

Roll No:	Name:	A

- Write your roll number and name. Nothing else is needed.

1. Consider the problem:

$$f^\star : \mathbb{R}^d \to \mathbb{R}, \quad y_i \approx f^\star(x_i)$$

Which statement best captures the learning objective?

(a) Minimize $\|x_i - y_i\|$

(b) Identify $f^\star$ exactly from data

(c) **Find $f_\theta \in \mathcal{F}$ minimizing empirical loss**

(d) Estimate probability density of inputs

2. Which statement correctly distinguishes approximation vs estimation error?

(a) Approximation depends on data, estimation on model class

(b) **Approximation depends on model class, estimation on data**

(c) Both depend only on optimization

(d) Both are identical

3. If $\sigma(z) = z$, then a multilayer network becomes:

(a) Nonlinear model

(b) Polynomial model

(c) **Linear (affine) model**

(d) Random mapping

4. Consider:

$$f(x) = \sigma(Wx + b), \quad W \in \mathbb{R}^{m \times d}$$

Which is true?

(a) Output dimension is d

(b) **Output dimension is** m

(c) Output dimension is scalar

(d) Depends only on σ

5. A shallow network:

$$f(x) = \sum_{j=1}^{m} a_j \sigma(w_j^T x + b_j)$$

Compared to deep networks, it:

(a) Always uses fewer parameters

(b) **May require exponentially many neurons**

(c) Is always more expressive

(d) Cannot approximate functions

6. Universal Approximation Theorem guarantees:

(a) Efficient learning

(b) Exact recovery of $f^\star$

(c) **Arbitrary approximation of continuous functions**

(d) Polynomial-time training

7. Which is NOT guaranteed by universal approximation?

(a) Existence of approximation

(b) Finite width suffices

(c) **Fast convergence rate**

(d) Approximation on compact sets

8. Depth improves representation because:

(a) Reduces parameters always

(b) Avoids nonlinearity

(c) **Enables compositional structure**

(d) Makes optimization convex

9. Which statement about ReLU is correct?

   (a) **Piecewise linear**

   (b) Saturates for large positive inputs

   (c) Always differentiable

   (d) Bounded

10. A neural network layer performs:

    (a) **Affine + nonlinear transformation**

    (b) Only linear mapping

    (c) Only nonlinear mapping

    (d) Random projection

11. Which best describes feature learning?

    (a) Removing noise

    (b) **Transforming representations across layers**

    (c) Reducing data size

    (d) Increasing parameters

12. The parameter space $\Theta \subset \mathbb{R}^p$ implies:

    (a) Finite hypothesis class

    (b) **Continuous parameterization**

(c) Discrete outputs

(d) No optimization needed

13. Which correctly defines empirical risk?

    (a) $\mathbb{E}[\ell]$ over distribution

    (b) **Loss over training samples**

    (c) Maximum loss

    (d) Gradient norm

14. Which component determines function class $\mathcal{F}$?

    (a) Dataset

    (b) Learning rate

    (c) Optimization method

    (d) **Architecture**

15. Why do deep networks represent some functions efficiently?

    (a) Linear structure

    (b) Small datasets

    (c) Convex loss

    (d) **Compositional hierarchy**

16. Increasing width mainly affects:

    (a) Depth

    (b) **Approximation capacity**

    (c) Optimization convexity

    (d) Data size

17. Which is true about affine maps?
    (a) **Linear + bias shift**
    (b) Always nonlinear
    (c) Cannot be composed
    (d) Not used in NN
18. Neural networks differ from classical methods because:
    (a) Fixed basis functions
    (b) **Adaptive basis functions**
    (c) No parameters
    (d) Deterministic outputs
19. Function approximation becomes difficult mainly due to:
    (a) Low dimension
    (b) Small datasets
    (c) **High-dimensional parameter space**
    (d) Convexity
20. The key tradeoff in neural networks is:
    (a) Speed vs memory
    (b) **Approximation vs estimation**
    (c) Depth vs width only
    (d) Data vs labels
21. Which statement best captures the role of depth in neural networks?
    (a) It reduces model complexity

(b) **It enables compositional representation of functions**

(c) It guarantees convex optimization

(d) It eliminates overfitting

22. Why are neural networks considered universal approximators?

(a) **They can approximate any continuous function arbitrarily well**

(b) They require no data

(c) They are always linear

(d) They can exactly represent any function

23. Which is the main limitation of increasing model capacity indefinitely?

(a) Reduced approximation ability

(b) Reduced optimization complexity

(c) Elimination of training error

(d) **Increased risk of overfitting and poor generalization**

24. What is the role of nonlinear activations in building complex decision boundaries?

(a) They simplify boundaries into linear forms

(b) **They allow the composition of nonlinear transformations**

(c) They reduce dimensionality

(d) They remove bias terms

25. Which statement best summarizes the learning problem in neural networks?

(a) It is purely a statistical estimation problem

(b) It is purely an optimization problem

(c) **It combines approximation, estimation, and optimization challenges**

(d) It is a deterministic mapping problem only